

Architect’s Brief

Fleet Maintenance Coordination System

1. Architecture Purpose

This brief prepares the Fleet Maintenance Coordination System for system architecture modeling. It identifies the likely system boundary, external participants, internal responsibilities, integrations, data ownership concerns, workflows, and quality drivers.

This brief is not a product requirements document and does not define implementation design or code structure.

2. System Boundary

The subject system is the **Fleet Maintenance Coordination System**.

The system owns vehicle maintenance coordination, alert creation, preventive maintenance evaluation, driver issue intake, work order lifecycle management, maintenance assignment, vehicle availability status, maintenance history, notifications, reporting, audit history, and integration status tracking.

The system does not own physical vehicles, telematics hardware, onboard diagnostic systems, vendor repair systems, parts warehouse operations, accounting, procurement, route dispatching, payroll, or vehicle control.

3. External Participants

Participant	Role in the Architecture
Fleet Manager	Reviews fleet health, maintenance priority, fleet availability, and operational impact.
Maintenance Coordinator	Creates, schedules, assigns, and monitors work orders.
Driver	Reports vehicle issues, submits inspection results, and receives maintenance instructions.
Internal Technician	Accepts, performs, and updates assigned maintenance work.
Repair Vendor	Accepts, performs, and updates assigned external repair work.
Operations Manager	Reviews downtime, availability, and maintenance impact on operations.
System Administrator	Configures vehicles, users, maintenance

Participant	Role in the Architecture
	rules, vendors, notifications, and integrations.
Telematics Device	Provides vehicle health, location, mileage, and usage telemetry.
Vehicle Diagnostic System	Provides diagnostic trouble codes and vehicle health events.
Vendor Management System	Optional external system for vendor work order exchange.
Notification Provider	Sends alerts, work assignments, reminders, and escalations.
Identity Provider	Authenticates users and resolves roles where configured.
Parts or Procurement System	Optional future system for parts availability or purchasing workflow.
Accounting System	Optional future system for vendor invoice or repair cost reconciliation.

4. Candidate Internal System Responsibilities

Responsibility Area	Description
Vehicle Registry Management	Manages vehicles, depots, regions, assignments, telematics associations, and lifecycle state.
Telematics Ingestion	Receives mileage, location, engine hours, and vehicle telemetry from telematics devices or gateways.
Diagnostic Event Processing	Interprets diagnostic events and maps them to maintenance severity or action categories.
Telemetry Freshness Monitoring	Identifies stale, missing, invalid, duplicated, or delayed vehicle data.
Preventive Maintenance Evaluation	Applies service interval rules to identify upcoming, due, and overdue maintenance.
Alert Management	Creates, prioritizes, routes, acknowledges, and resolves maintenance alerts.
Driver Issue Management	Captures driver-reported issues, safety status, inspection failures, and attachments.
Work Order Management	Creates, assigns, tracks, updates, completes, and cancels maintenance work

Responsibility Area	Description
	orders.
Assignment Coordination	Coordinates work between internal technicians and external vendors.
Vendor Coordination	Manages vendor assignment, acceptance, rejection, status updates, and delayed work.
Vehicle Availability Management	Tracks whether vehicles are available, in service, assigned to maintenance, or out of service.
Notification Management	Sends maintenance alerts, reminders, escalations, and work updates.
Reporting and Analytics	Provides fleet availability, compliance, downtime, vendor performance, and work order reporting.
Audit Logging	Records maintenance, assignment, work order, availability, configuration, and integration events.
Integration Monitoring	Tracks telematics, diagnostic, vendor, notification, identity, and optional procurement integration failures.

5. Candidate Data Stores

Data Store	Architectural Purpose
Vehicle Registry Store	Stores vehicles, identifiers, fleet grouping, depots, assignments, and lifecycle state.
Telematics Store	Stores received vehicle telemetry, mileage, engine hours, location, and freshness metadata.
Diagnostic Event Store	Stores fault codes, diagnostic events, severity mapping, and source details.
Preventive Maintenance Rule Store	Stores maintenance schedules, vehicle class rules, service intervals, and due thresholds.
Maintenance Alert Store	Stores alerts, severity, source, status, assignment, and resolution details.
Driver Issue Store	Stores driver reports, safety status, inspection results, comments, and attachments.
Work Order Store	Stores work orders, tasks, assignees,

Data Store	Architectural Purpose
	status, priority, due dates, and completion summaries.
Vendor Store	Stores vendor profiles, service categories, regions, assignment preferences, and contact details.
Vehicle Availability Store	Stores current and historical availability state for each vehicle.
Maintenance History Store	Stores completed work, notes, attachments, downtime, and historical service records.
Notification Store	Stores notification requests, recipients, delivery status, failures, and templates.
Reporting Store	Stores aggregates for fleet availability, downtime, compliance, vendor performance, and work aging.
Integration Status Store	Stores external request status, retry state, failures, and correlation references.
Audit Log Store	Stores significant user, system, maintenance, and integration events.

6. Candidate External Interfaces

Interface	Provider	Consumer	Notes
Vehicle Telemetry Ingestion Interface	Fleet Maintenance Coordination System	Telematics devices or telematics gateway	Receives mileage, engine hours, location, and vehicle health telemetry.
Diagnostic Event Interface	Vehicle Diagnostic System	Fleet Maintenance Coordination System	Provides diagnostic trouble codes and vehicle health events.
Driver Issue Reporting Interface	Fleet Maintenance Coordination System	Driver mobile experience	Supports issue reports, safety status, inspection results, and attachments.
Fleet Management Interface	Fleet Maintenance Coordination System	Fleet manager and coordinator experience	Supports vehicles, alerts, work orders, assignments, and availability.

Interface	Provider	Consumer	Notes
Technician Work Order Interface	Fleet Maintenance Coordination System	Internal technician experience	Supports assigned work order review, acceptance, update, and completion.
Vendor Work Order Interface	Fleet Maintenance Coordination System	Repair vendor portal or vendor system	Supports vendor assignment, acceptance, rejection, and status updates.
Notification Delivery Interface	Notification Provider	Fleet Maintenance Coordination System	Sends maintenance alerts, reminders, escalations, and work updates.
User Authentication Interface	Identity Provider	Fleet Maintenance Coordination System	Authenticates fleet staff, technicians, vendors, and administrators where configured.
Parts Availability Interface	Parts or Procurement System	Fleet Maintenance Coordination System	Optional future interface for parts availability or purchase workflow.
Cost Reconciliation Interface	Accounting System	Fleet Maintenance Coordination System	Optional future interface for repair cost or invoice reconciliation.

7. Primary Workflows for Architecture Modeling

Workflow	Summary
Ingest Vehicle Telemetry	Receive, validate, store, and expose vehicle telemetry and freshness status.
Process Diagnostic Event	Convert diagnostic trouble code or health event into alert or work order candidate.
Submit Driver Issue	Driver reports a vehicle problem, safety status, inspection result, and optional attachments.
Evaluate Preventive Maintenance	System checks vehicles against configured maintenance intervals and identifies due work.
Create Maintenance Alert	System creates alert from telemetry,

Workflow	Summary
	diagnostic event, driver report, or preventive rule.
Create Work Order	Coordinator creates work order from alert, schedule, manual request, or inspection result.
Assign Work Order	Coordinator assigns work to internal technician or external vendor.
Accept or Reject Assignment	Technician or vendor accepts, rejects, or requests clarification.
Update Work Order Status	Assignee updates progress, blockers, parts status, completion notes, or rejection.
Return Vehicle to Service	Completed work causes vehicle availability to be restored after review or completion.
Escalate Overdue Work	System identifies overdue work order or stalled vendor response and notifies responsible users.
Report Fleet Availability	System summarizes maintenance impact on usable fleet capacity.

8. Key State Models

Vehicle Availability State

Candidate states:

1. Available.
2. In service.
3. Assigned for maintenance.
4. Out of service.
5. Awaiting inspection.
6. Waiting on parts.
7. Awaiting vendor update.
8. Ready for pickup.
9. Retired.

Maintenance Alert State

Candidate states:

1. New.
2. Notified.
3. Acknowledged.
4. Linked to work order.

5. Dismissed.
6. Resolved.
7. Escalated.

Work Order State

Candidate states:

1. Draft.
2. Open.
3. Assigned.
4. Accepted.
5. Rejected.
6. In progress.
7. Waiting on parts.
8. Waiting on approval.
9. Completed.
10. Canceled.
11. Reopened.

Preventive Maintenance State

Candidate states:

1. Not due.
2. Due soon.
3. Due.
4. Overdue.
5. Scheduled.
6. In service.
7. Completed.
8. Deferred.

Integration Attempt State

Candidate states:

1. Pending.
2. Sent.
3. Succeeded.
4. Failed.
5. Retrying.
6. Timed out.
7. Requires manual review.
8. Abandoned.

9. Key Quality Drivers

Quality Driver	Architectural Implication
Reliability	Maintenance state must survive temporary telematics, vendor, and notification failures.
Data Freshness	Vehicle health views must distinguish current telemetry from stale or missing data.
Auditability	Alerts, assignments, status changes, safety decisions, and vehicle availability changes require traceable history.
Operational Visibility	Fleet managers need clear visibility into downtime, overdue work, vendor delays, and alert backlogs.
Mobile Usability	Driver and technician workflows should support quick mobile interaction.
Security	Drivers, technicians, vendors, managers, and administrators require distinct access boundaries.
Vendor Isolation	Vendors should only see work orders assigned to them.
Configurability	Maintenance intervals, severity thresholds, vehicle group rules, notifications, and vendor policies vary by fleet.
Scalability	The system must handle growth in vehicles, telemetry volume, work orders, vendors, and regions.
Resilience	Telemetry and work order updates should support retry and reconciliation after temporary failures.

10. Architecture Risks

Risk	Architectural Concern
Diagnostic ambiguity	Fault codes may not map cleanly to maintenance severity or action.
Stale telemetry	Maintenance decisions may be wrong if telemetry freshness is not visible.
Vehicle safety authority	The system must clearly model who can mark a vehicle unsafe or return it to service.
Vendor update gaps	External vendors may fail to update status

Risk	Architectural Concern
Work order duplication	promptly, creating operational uncertainty. Driver reports, diagnostic events, and preventive rules may create duplicate work candidates.
Preventive maintenance rule complexity	Rules may vary by vehicle type, region, manufacturer, and fleet policy.
Offline field usage	Drivers or technicians may need low-connectivity workflows.
Attachment handling	Photos and repair documents may create storage, privacy, and retention concerns.
Availability consistency	Vehicle availability must remain aligned with work order state and operational needs.
Integration variability	Telematics and vendor systems may use different data formats and delivery patterns.

11. Open Architecture Questions

1. Which telematics providers and data formats must be supported first?
2. Are diagnostic events received directly from vehicles, through telematics providers, or through another gateway?
3. How should diagnostic trouble codes be mapped to severity and recommended action?
4. Who can mark a vehicle unsafe to operate?
5. Who can return a vehicle to service after maintenance?
6. Should driver issue reports automatically create alerts, work orders, or review tasks?
7. How should duplicate alerts and work order candidates be detected and merged?
8. Should vendors update work through a portal, email-based workflow, or system integration?
9. What vehicle availability states are required for operations planning?
10. Should preventive maintenance be based on mileage, engine hours, calendar time, usage pattern, or combined rules?
11. What offline capabilities are required for drivers or technicians?
12. What attachment types and size limits are required?
13. How should maintenance records be retained for compliance, resale, or audit?
14. Which reports require real-time data versus historical aggregation?
15. What integration failures should trigger user alerts versus administrative review?

12. Recommended Modeling Scope

For the initial architecture model, focus on:

1. Vehicle registry and availability tracking.
2. Driver issue reporting.
3. Telematics and diagnostic event ingestion.
4. Maintenance alert creation.
5. Preventive maintenance evaluation.
6. Work order creation and assignment.
7. Technician and vendor work updates.
8. Notification delivery.
9. Maintenance history.
10. Audit and integration status tracking.
11. Fleet availability reporting.

Defer advanced modeling for:

1. Vendor system integration.
2. Parts inventory integration.
3. Cost tracking and accounting integration.
4. Predictive maintenance modeling.
5. Offline mobile synchronization.
6. Automated vendor selection.
7. Procurement workflows.